

RIZHAO, China

WMO: 549450

Lat: 35.4681N

Lon: 119.5550E

Elev: 37

StdP: 100.88

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.9	(c) -5.3	(d) -21.6	(e) 0.5	(f) -2.1	(g) -19.4	(h) 0.7	(i) -2.3	(j) 10.1	(k) -1.7	(l) 9.1	(m) -0.2	(n) 3.0	(o) 320	(p) 0.399

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 4.7	(c) 31.7	(d) 25.4	(e) 30.3	(f) 25.1	(g) 29.2	(h) 24.8	(i) 27.6	(j) 30.1	(k) 27.0	(l) 29.1	(m) 26.4	(n) 28.2	(o) 3.1	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.9	22.7	29.1	26.3	21.9	28.5	25.8	21.2	28.0	88.4	30.2	85.5	29.1	82.9	28.2	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.8	7.4	6.4	-9.3	34.5	2.0	2.0	-10.7	35.9	-11.9	37.0	-13.0	38.1	-14.4	39.6		
			-11.1	28.6	1.8	0.6	-12.4	29.0	-13.5	29.4	-14.5	29.7	-15.9	30.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.0	0.6	2.4	7.2	13.0	18.4	22.0	25.8	26.0	22.3	16.9	9.5	3.0
	DBStd	9.40	3.11	3.16	3.57	3.40	3.19	2.48	2.21	2.06	2.31	3.06	4.01	3.43
	HDD10.0	892	293	212	103	9	0	0	0	0	0	1	57	218
	HDD18.3	2353	551	445	346	166	38	2	0	0	2	63	266	476
	CDD10.0	2346	0	0	16	98	261	361	489	497	369	213	41	1
	CDD18.3	766	0	0	0	4	41	113	231	239	121	18	0	0
	CDH23.3	5373	0	0	1	34	239	497	1904	2141	531	26	0	0
	CDH26.7	1254	0	0	0	9	61	120	489	512	62	1	0	0
Wind	WSAvg	2.9	3.0	3.0	3.1	3.2	2.9	2.5	2.4	2.7	2.8	2.8	3.0	3.0
Precipitation	PrecAvg	799	10	21	22	41	81	90	188	169	88	38	34	16
	PrecMax	1223	48	60	70	110	246	207	434	367	304	166	212	55
	PrecMin	516	0	2	1	6	2	4	71	44	5	0	0	0
	PrecStd	203	13	16	19	25	54	55	85	87	78	44	49	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.9	13.0	21.4	27.2	31.7	32.3	33.9	33.2	30.4	25.9	20.1	13.4
		MCWB	5.0	7.0	12.3	16.6	18.2	21.0	27.5	26.7	23.1	18.0	13.7	9.9
	2%	DB	7.6	10.2	17.7	22.9	28.0	29.7	31.8	31.2	28.2	23.8	18.0	11.2
		MCWB	4.4	5.2	10.4	14.7	17.5	21.5	26.7	26.4	22.4	17.4	13.6	8.3
	5%	DB	6.4	8.4	14.8	20.4	25.9	27.7	30.4	30.0	27.0	22.6	16.8	9.6
		MCWB	3.5	4.6	8.9	13.4	16.8	21.6	26.1	25.9	21.9	17.4	13.0	6.5
	10%	DB	5.2	7.1	12.4	18.3	23.9	26.0	29.2	29.1	26.0	21.5	15.5	8.2
		MCWB	2.6	4.0	7.4	12.2	16.7	20.8	25.8	25.5	21.2	17.0	11.8	5.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	8.4	13.8	18.5	21.1	25.8	28.5	28.5	26.4	21.0	16.7	11.4
		MCDB	8.5	10.9	19.6	24.2	26.2	29.0	32.0	31.0	28.3	22.8	17.6	12.1
	2%	WB	5.5	6.8	11.5	16.0	19.7	24.2	27.6	27.7	24.6	19.8	15.3	9.0
		MCDB	6.8	8.7	15.8	21.1	24.5	26.9	30.5	29.9	26.6	22.1	17.2	10.5
	5%	WB	4.2	5.7	9.8	14.5	18.8	23.0	27.0	27.2	23.6	19.0	13.8	7.2
		MCDB	5.8	7.4	13.3	18.4	23.2	25.4	29.4	29.1	25.6	21.5	16.0	9.2
	10%	WB	2.9	4.6	8.4	13.4	17.9	22.0	26.3	26.7	22.6	17.9	12.2	5.5
		MCDB	4.9	6.5	11.5	16.5	21.7	24.1	28.4	28.4	25.0	20.8	14.9	7.7
Mean Daily Temperature Range	5% DB	MDBR	6.3	6.2	6.7	6.9	6.8	5.2	4.5	4.7	5.9	6.6	6.7	6.4
		MCDBR	7.4	7.9	10.5	10.7	10.9	8.6	6.5	5.7	6.8	7.4	7.5	7.4
		MCWBR	5.4	5.4	5.9	5.7	4.8	3.9	3.0	2.9	3.4	4.3	5.4	5.7
	5% WB	MCDDBR	6.4	6.4	9.1	8.9	8.5	6.1	5.4	4.5	5.1	5.8	6.4	6.4
		MCWBR	5.4	5.0	5.7	5.4	4.6	3.3	2.9	2.5	3.3	4.4	5.5	5.7
Clear-Sky Solar Irradiance	taub	0.523	0.585	0.624	0.634	0.651	0.757	0.733	0.679	0.602	0.627	0.589	0.526	
	taud	1.753	1.662	1.614	1.642	1.647	1.501	1.601	1.692	1.791	1.660	1.684	1.763	
	Ebn at Noon	621	636	660	687	683	613	624	647	671	592	555	583	
	Edh at Noon	176	216	246	251	252	292	262	235	203	213	188	165	
All-Sky Solar Radiation	RadAvg	2.50	3.09	4.37	5.12	5.52	5.20	4.65	4.59	4.17	3.53	2.62	2.26	
	RadStd	0.28	0.40	0.35	0.36	0.43	0.41	0.50	0.53	0.37	0.38	0.33	0.21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+62	N/A

Nomenclature: See separate page